13_Positive-Negative Definite & Semi-Definite Matrix

Solved Eigenvalues

L

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Step 1: Subtract λI

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{bmatrix}$$

Step 2: Determinant

$$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - (1)(2)$$

Expand:

$$\begin{aligned} &= 12 - 4\lambda - 3\lambda + \lambda^2 - 2 \\ &= \lambda^2 - 7\lambda + 10 \end{aligned}$$

Step 3: Solve polynomial

$$\lambda^2 - 7\lambda + 10 = 0$$

Factor:

$$(\lambda - 5)(\lambda - 2) = 0$$

✅ **Eigenvalues**

$$\boxed{\lambda_1 = 5, \lambda_2 = 2}$$

Positive Definite

1. Positive Definite (PD)

Definition

A symmetric matrix A is **positive definite** if

$$x^\top A x > 0 \text{for all } x \neq 0$$

Notation

$$A \succ 0$$

How to recognize it (any ONE is enough)

- All **eigenvalues > 0**
- All **leading principal minors > 0**
- **Cholesky decomposition exists**:

$$A = LL^\top$$

Intuition

The quadratic form is **strictly positive in every direction**.

Example

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

- Eigenvalues: $2,3 > 0$
- $x^\top A x = 2x_1^2 + 3x_2^2 > 0$

✅ **Positive definite**

2. Positive Semi-Definite (PSD)

Definition

$$x^\top A x \geq 0 \text{for all } x$$

Notation

$$A \succeq 0$$

Equivalent conditions

- All **eigenvalues ≥ 0**
- Can be written as

$$A = B^\top B$$

Intuition

The quadratic form is **never negative**, but may be zero in some directions.

Example

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

- Eigenvalues: 2,0
- $x^\top A x = (x_1 + x_2)^2 \geq 0$

✅ **Positive semi-definite**
❌ Not PD (because one eigenvalue is 0)

3. Negative Definite (ND)

Definition

$$x^\top Ax < 0 \text{for all } x \neq 0$$

Notation

$$A \prec 0$$

Equivalent conditions

- All **eigenvalues < 0**
- $-A$is **positive definite**

Intuition

The quadratic form is **strictly negative in every direction**.

Example

$$A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}$$

- $x^\top Ax = -2x_1^2 - x_2^2 < 0$

✅ **Negative definite**

4. Negative Semi-Definite (NSD)

Definition

$$x^\top Ax \leq 0 \text{for all } x$$

Equivalent condition

- All **eigenvalues ≤ 0**

Intuition

The quadratic form is **never positive**, but may be zero in some directions.

Example

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$$

- Eigenvalues: $-1, 0$

✅ **Negative semi-definite**

8. Summary Table

Matrix Type	Condition on $x^\top Ax$	Eigenvalues
Positive Definite	> 0	$\lambda_i > 0$
Positive Semi-Definite	≥ 0	$\lambda_i \geq 0$
Negative Definite	<0	$\lambda_i < 0$
Negative Semi-Definite	≤ 0	$\lambda_i \leq 0$
Indefinite	mixed	mixed signs